Relation-Aware High-Order Interaction Network for Remote Sensing Image Change Detection

Huihui Dong, Xinyu Du, Zhijie Li, Zongfang Ma, Yumeng Wang, Hao Zhu, *Member, IEEE*, and Wenping Ma, *Senior Member, IEEE*

***Abstract*—Remote sensing image change detection plays a vital role in ecological monitoring, urban planning, and infrastructure assessment. However, accurately identifying changes in high-resolution remote sensing imagery remains challenging due to multitemporal imaging variations and complex backgrounds. A key limitation of existing methods is their inadequate modeling of spatiotemporal relationships, which restricts their ability to distinguish meaningful changes from irrelevant variations. To alleviate this limitation, we propose a relation-aware high-order spatial interaction network (RHighNet), which enhances spatiotemporal feature learning through spatial, channel, and high-order relation modeling across bitemporal images. The proposed RHighNet consists of two key components: 1) a temporal relationship-aware module (TRAM) that captures bitemporal dependencies via a dual-branch attention mechanism, effectively highlighting change-related features in both spatial and channel dimensions and 2) a high-order spatiotemporal interaction module (HOSTIM) that employs recursive gated convolutions to model higher order feature interactions, enabling robust representation of complex changes beyond two-order interactions in self-attention. In addition, a multistage loss optimization strategy is introduced to stabilize training and refine feature learning progressively. Extensive experiments on four benchmark datasetsłLEVIR-CD, WHU-CD, SYSU, and CDDłdemonstrate the effectiveness and superiority of our method in remote sensing image change detection. Code available at https://github.com/yizhilanmaodhh/RHighNet**

***Index Terms*—Change detection, high-order spatial interaction, relation attention mechanisms, remote sensing images.**

I. Introduction

REMOTE sensing image change detection aims to identify and analyze altered regions by comparing surface images acquired at different times. With the rapid advancements in remote sensing technology, change detection has gained increasing prominence in various fields, including ecological environment protection [1], urban planning [2], and resource management [3]. Given the dynamic nature of surface environments, change detection serves as an effective monitoring tool for rapidly acquiring information about surface modifications, thereby supporting scientific decision-making.

Traditional change detection methods for remote sensing images can be broadly categorized into pixel-based, feature-based, and model-based approaches [4]. Pixel-based methods directly compare the spectral differences between corresponding pixels in two images. Commonly used difference generation algorithms include simple differencing, ratio methods, and vegetation index comparison etc. These methods then apply threshold-based algebraic analysis to obtain change information. Change-vector analysis (CVA) [5] is a widely used method for extracting difference information, and with many improved variants, such as compressed CVA, robust CVA [6], and logical validation CVA [7]. However, these methods are highly susceptible to noise, illumination variations, and sensor inconsistencies, often resulting in a high number of false positives. Feature-based methods extract specific features from images, such as texture, shape, and principal components, before comparing them to identify change regions. Common techniques in this category include principal component analysis (PCA) [8], multivariate alteration detection (MAD) [9], and slow feature analysis (SFA) [10], etc. However, these feature extraction methods are often complex and may fail to fully encapsulate the semantic change information between bitemporal images. Model-based approaches consider spatial correlations between image pixels to improve detection accuracy, such as Markov random fields [11]. However, these models are challenging to construct and parameterize, making them inefficient for processing large-scale remote sensing image data.

In recent years, advancements in deep learning have introduced new opportunities for remote sensing image processing [12], significantly enhancing the accuracy and efficiency of change detection. By leveraging the powerful feature extraction capabilities of deep neural networks, these models can autonomously learn complex distributions and transformation patterns of image features. As a result, they exhibit strong adaptability and generalization capabilities, offering more precise support for environmental monitoring and emergency response. Most deep learning-based change detection algorithms are built upon U-Net architecture, which integrates temporal and spatial feature extraction, multiscale decoding, and change region recognition within a unified end-to-end

Received 4 May 2025; revised 2 August 2025; accepted 27 August 2025. Date of publication 1 September 2025; date of current version 8 September 2025. This work was supported in part by the Project of Collaborative Innovation Center of Shaanxi Provincial Department of Education under Grant 23JY038 and in part by the Natural Science Basic Research Project of Shaanxi Province under Grant 2024JC-YBQN-0706. *(Corresponding author: Huihui Dong.)*

Huihui Dong, Xinyu Du, Zhijie Li, Zongfang Ma, and Yumeng Wang are with the School of Information and Control Engineering, Xi'an University of Architecture and Technology, Xi'an 710055, China (e-mail: hhdongxd@163.com; xydududu@163.com; lizhijie@xauat.edu.cn).

Hao Zhu and Wenping Ma are with the Key Laboratory of Intelligent Perception and Image Understanding of Ministry of Education, School of Artificial Intelligence, Xidian University, Xi'an, Shaanxi 710071, China.

Digital Object Identifier 10.1109/TGRS.2025.3604400

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:07:29 UTC from IEEE Xplore. Restrictions apply.

framework [13]. For temporal image pairs, Siamese-network encoders are commonly employed to handle dual input features. These approaches formulate change detection as a binary segmentation problem, utilizing a variety of deep encoders and decoders, including convolutional neural networks (CNNs) [14] (e.g., ResNet, DenseNet, and EfficientNet), graph neural networks (GNNs) [15], recurrent neural networks [16], attention mechanisms, and transformer architectures (e.g., Vision Transformer and Swin Transformer), etc.

Furthermore, hybrid network architectures that integrate the advantages of multiple models have been proposed to better represent and aggregate temporal image features, thereby improving detection performance. For example, some methods combine CNN with Transformer [17], [18], using CNN to extract multiscale spatial features and Transformer to model long-distance dependencies. This combination enhances the model's ability to jointly perceive both local details and global context in complex scenes. In addition, some methods combine CNN with GNN [19], where GNN is used to construct spatial neighborhood maps or heterogeneous maps of images. Benefit from the modeling of the spatial relationships between pixels and interactions of local regions, which help to capture subtle changes and the evolution of local structures in image pairs. These hybrid networks are effective for improving the accuracy of change detection in remote sensing images by leveraging the complementary advantages of different models.

Despite these advances, precise change detection remains challenging due to factors such as task-irrelevant disturbances (e.g., seasonal changes, weather differences, and building renovations) and increasingly complex geographical information in high-resolution remote sensing images [20], [21]. These are typical features of bitemporal images, which exacerbate the complexity of temporal feature relationships, limiting the accuracy and adaptability of the change detection model to a certain extent.

To alleviate these issues, in this article, we propose a relation-aware high-order spatial interaction network (RHighNet), designed to extract more discriminative spatiotemporal features by modeling the intricate association information between bitemporal images, thereby improving change detection performance. Specifically, temporal image pairs are first processed through convolutional encoders to extract feature representations. These encoded features are then fed into a high-order spatio-temporal interaction module (HOSTIM), which models complex high-order spatiotemporal dependencies. Compared with two-order interactions in self-attention, HOSTIM can improve the ability to deal with high-order complex change relations by extending to higher orders, so as that improve the robustness in complex change scenes. Then, a temporal relationship-aware module (TRAM) is proposed to effectively capture spatiotemporal dependencies at different feature levels across both channel and spatial dimensions. The spatiotemporal features aggregated at each layer with different scales are subsequently passed through a progressively cascaded decoder, which incrementally restores spatial resolution to facilitate pixel-level predictions. Finally, to further optimize network performance, we introduce a structure loss function, which is combined with conventional cross-entropy loss, enables multistage optimization, thereby enhancing the generalization capabilities of the model and further improving the detection performance.

The main contributions of our method are summarized as follows.

1) We propose a RHighNet, which integrates CNN and higher order interaction networks within a U-shaped architecture for remote sensing image change detection. This framework effectively models temporal image features and captures the complex dependencies between them.
2) To adaptively model important spatiotemporal relationships between bitemporal images, we propose the TRAM that combines channelwise and spatialwise attention mechanisms. This design captures multiscale feature dependencies across different dimensions, enhancing the ability to discern key spatiotemporal patterns for the model.
3) To address the complex higher order spatiotemporal relationships between bitemporal images, we propose the HOSTIM, which extends spatiotemporal feature interactions to arbitrary orders, enabling higher order associations among features. By capturing more intricate spatiotemporal dependencies, the HOSTIM enhances the model's ability to interpret and process complex change dynamics, leading to more robust and accurate analysis.
4) Extensive experiments on four datasets (LEVIR-CD, WHU-CD, SYSU-CD, and CDD) demonstrate state-of-the-art performance of the proposed model, with ablation studies validating the efficacy of each component.

The remainder of this article is structured as follows. In Section II, we briefly review the studies related to our work. Section III describes the overall framework and the details of each part of our proposed network. The experimental results and the corresponding analyses are reported in Section IV. Finally, we summarize our contributions and conclude the article in Section V.

II. Related Work

A. Change Detection Based on CNN and Attention Mechanism

Over the past few decades, CNNs have been extensively utilized in change detection tasks due to their superior ability to extract local features. Most existing approaches employ a U-shaped encoder–decoder architecture for feature extraction, where the encoder typically follows either an early fusion or late fusion strategy. In early fusion, bitemporal images are concatenated as input, whereas in late fusion, features are extracted from two images separately. Based on whether the encoder weights are shared, late fusion methods can be further categorized into Siamese and pseudo-Siamese structures. For instance, Xie et al. [22] adopted a Siamese-structure encoder for feature extraction and introduced three fully convolutional networks with different fusion strategies and skip connections to perform change detection. Over time, encoder-based Siamese networks have become the mainstream approach, partly because they preserve the distinctive features of each

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:07:29 UTC from IEEE Xplore. Restrictions apply.

image while simultaneously capturing different information, thereby enhancing change detection accuracy.

Building on dual encoder architectures, Li et al. [21] proposed a Siamese NestedUNet method, which strengthens shallow features containing high-resolution and fine-grained details through dense connections, improving the detection of fine object boundaries and small change targets. Similarly, Li et al. [20] developed a dense attention refinement network that employs dense connections in both the encoder and decoder to aggregate multilevel features while incorporating temporal context information through a hybrid attention mechanism. To further enhance the robustness and accuracy of change detection models, various attention mechanisms have been integrated into Siamese networks [23]. Rather than merely increasing feature dimensions, these mechanisms reweight features by computing attention scores for each position or channel, enabling the network to focus more effectively on essential information. Commonly used attention mechanisms include channel attention [24], spatial attention [25], and cross-attention [26] strategies. To mitigate the influence of false change information, Chen et al. [27] introduced a dual attentive fully convolutional Siamese network that captures long-range dependencies by a dual-attention mechanism, thereby enhancing the discriminative power of extracted features. In [28], researchers proposed a dual-branch multilevel intertemporal network, which efficiently derives change representations by integrating self-attention (SelfAtt) and cross-attention (CrossAtt) within a unified module. This design incorporates a cross-temporal joint attention (JointAtt) block, which guides the global feature distribution of each input while coupling information across different layers, effectively suppressing task-irrelevant noise. In addition, Liu et al. [29] developed a multiscale interaction network that enhances temporal feature interactions through a joint cross-attention module, multiscale offset calibration module, and feature integration module, further improving detection performance.

These methods leverage attention mechanisms to highlight features relevant to change regions, significantly enhancing the effectiveness of change detection. However, most of the methods pay attention to two branches separately, and only use strategies such as subtraction, addition, or cascade to aggregate bitemporal features, with limited interaction between bitemporal images. Drawing inspiration from the principle that attention mechanisms emphasize crucial feature information, and recognizing the complementary nature of spatial and channel information in feature fusion [30], we propose a TRAM to explore more fully temporal associations in terms of spatial and channel dimensions by attention interaction. As a result, more attention is focused on changing information between bitemporal images rather than assuming that their contributions are equal, as in usual feature fusion, thus improving the detection accuracy and robustness.

B. Change Detection Based on Transformer

Transformer-based architectures significantly challenge the long-standing dominance of CNNs. Based on self-attention mechanisms, Vision Transformers have achieved state-of-the-art performance in a wide range of visual tasks, including image classification [31], object detection [32], semantic segmentation [33], and video understanding [34]. In the field of change detection, Transformer-based models are often incorporated with CNNs, either in a parallel or serial structure, to harness the strengths of both architectures and enhance detection performance [35], [36], [37]. Chen et al. [38] introduced the Vision Transformer into change detection, named as the bitemporal image transformer (BIT), which first employs several convolutional blocks to extract local features from each input image, and then feeds them into BIT to extract global features. Feng et al. [39] proposed an intrascale cross-interaction and interscale feature fusion network, explicitly tapping the potential of integrating CNN and transformer so that offering a more powerful detection capability. Song et al. [35] proposed a transformer network based on dual-feature mixed attention, which applies a dual-feature mixed attention module to fuse fine and coarse features. It not only extracts more specific regions of interest but also overcomes misjudgments from oversampling and synchronizes feature extraction with target information integration. Wu et al. [40] introduced a cross Swin Transformer-based Siamese U-Net, which leverages the Swin Transformer's strong local and global feature-capturing ability to enhance the extraction of rich and effective features from remote sensing images. To further refine the integration of CNNs and transformers, Zhao et al. [41] proposed a geometric representation transformer. This model consists of a geometry-based Swin Transformer encoder, a Laplacian attention fusion module, and a UNet++CD decoder. By embedding geometric priors within the transformer structure, this approach achieves more accurate detection of detailed change information. Furthermore, other methods have integrated attention mechanisms [42], GNNs [43], and LSTMs [44], etc., into Transformer-based models to further improve detection performance.

Unlike these approaches, our method integrates CNNs with higher order spatial interaction networks to explore their combined effectiveness in remote sensing image change detection. Compared with transformers, higher order spatial interaction networks preserve key advantages inherent in dot-product operations, including adaptability, global context modeling ability, and higher order spatial interaction capability. These properties facilitate more sophisticated modeling of high-order dependencies while maintaining computational efficiency. By incorporating a higher order spatial interaction module, the proposed model captures complex higher order relationships between temporal features, leading to further improvements in detection accuracy and robustness.

III. Methodology

A. Overall Architecture

Fig. 1 illustrates the overall framework of the proposed RHighNet. Specifically, the model comprises two parallel branches.

1) *Single-Temporal Feature Branch:* Independently propagates single-temporal image features to preserve original

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:07:29 UTC from IEEE Xplore. Restrictions apply.

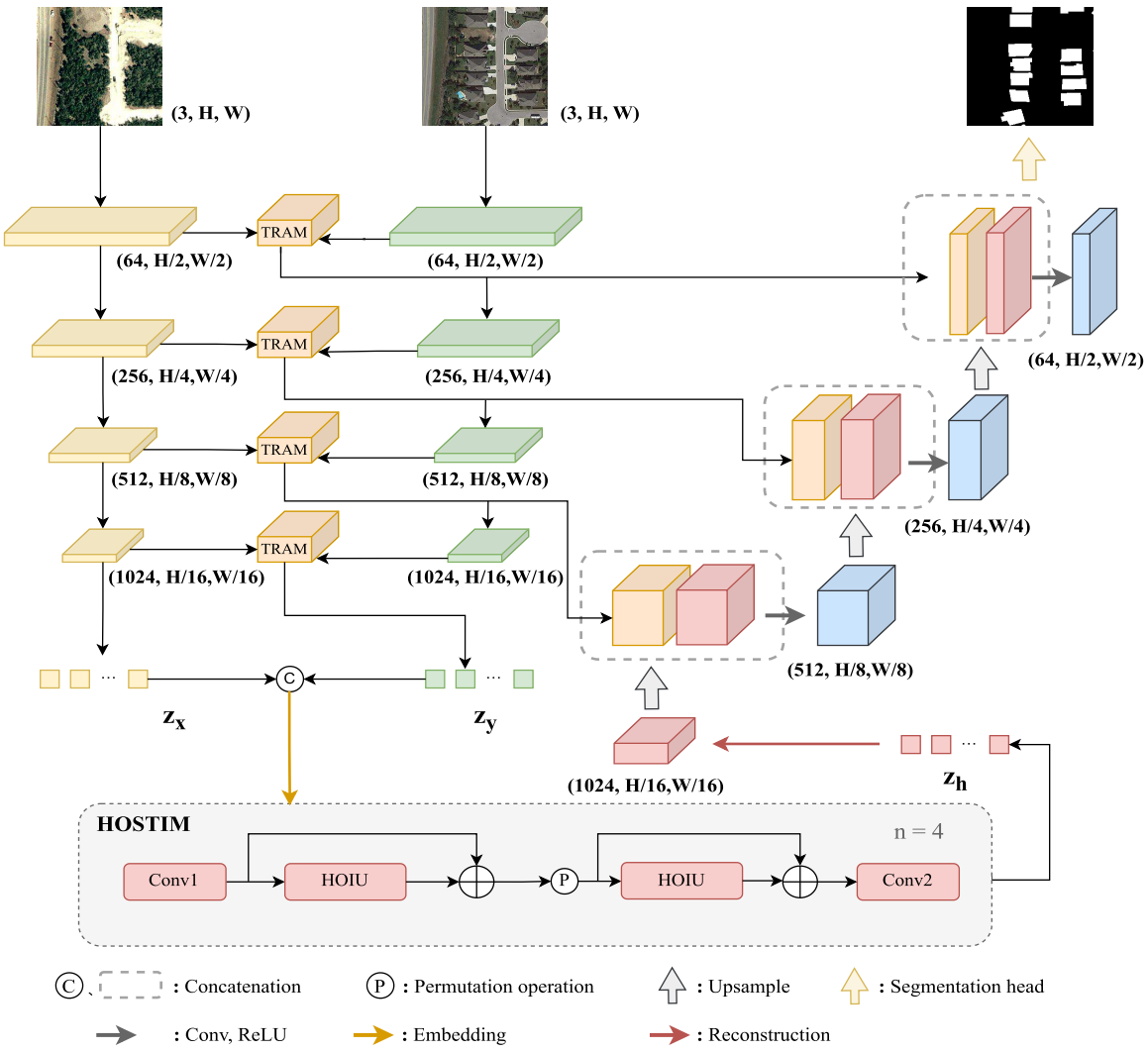


Fig. 1. Overview of the RHighNet. RHighNet preserves the original spatiotemporal information by a single-temporal feature branch, captures change features by a bitemporal feature fusion branch, and gradually refines high-order features by HOSTIM, and finally generates change detection results by the decoder with an upsampler.

spatiotemporal information and prevent over-smoothing of critical temporal features during fusion.

2) *Bitemporal Feature Fusion Branch:* Focuses on interactive fusion to enhance different information, and improving sensitivity to change regions by capturing the dependence relationship between bitemporal features.

In the shallow stage, each CNN block is followed by a TRAM, which facilitates the initial interaction and fusion of bitemporal features at an early stage. In the deeper stages, the model combines the features from both the single-temporal feature branch and bitemporal feature fusion branch to generate joint features that have rich spatiotemporal information. These combined features are then fed into the HOSTIM. HOSTIM performs higher order feature interactions through a recursive mechanism to gradually enhance the representation of complex features. Its core component, the higher order interaction block (HOIB), is able to model complex neighborhood relationships layer by layer along temporal and channel dimensions and capture finer spatiotemporal dependency information.

Finally, the features from the bitemporal fusion branch are concatenated with the high-order interaction features output by HOSTIM, are fed into the decoder. The spatial resolution is progressively restored through a cascade upsampler within the decoder, ultimately generating the change detection results.

B. Temporal Relation-Aware Module

In the change detection task, accurately modeling the relationships between spatiotemporal features is crucial for improving detection accuracy. Unlike common fusion methods that treat bitemporal features equally, the proposed TRAM dynamically prioritizes change-related regions, directing greater attention to discriminative spatiotemporal patterns from spatial and channel dimensions. The structure of TRAM is shown in Fig. 2.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:07:29 UTC from IEEE Xplore. Restrictions apply.

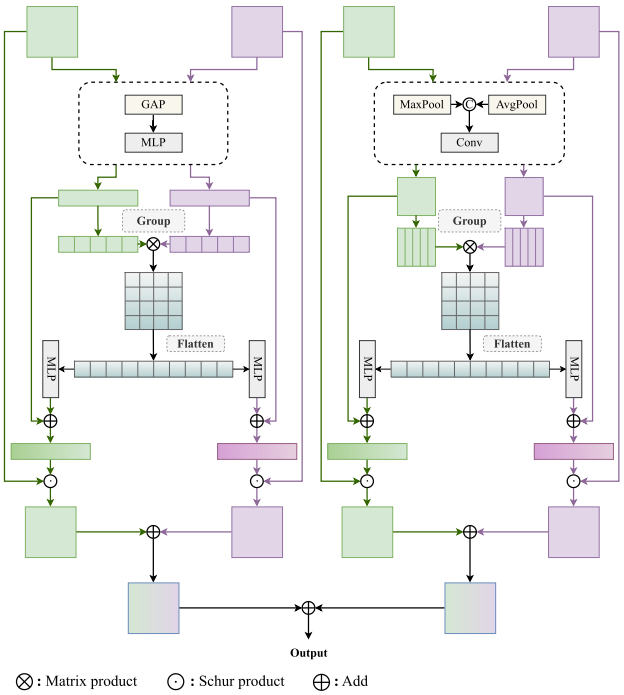


Fig. 2. Structure of TRAM. (Left) Channel relation attention. (Right) Spatial relation attention. TRAM enhances the extraction of change features through spatial and channel attention relationship modeling.

1) Spatial Relationship Modeling: For the given feature pairs $x_1 \in \mathbb{R}^{C\times H\times W}$ and $x_2 \in \mathbb{R}^{C\times H\times W}$ from two branches, respectively, the spatial attention A_s is first calculated as

$$A_s = \sigma(\text{Conv}(\text{Concat}(\text{MaxPool}(x), \text{AvgPool}(x)))) \quad (1)$$

where MaxPool(·) and AvgPool(·) represent global max pooling and global average pooling operations, Concat(·) represents concatenation along channel dimension, Conv(·) represents convolution operation, and $\sigma(\cdot)$ denotes sigmoid activation function. The spatial attention enhances the model to focus on key spatial regions and improves feature representation.

To better model the spatial relationship between bitemporal input features, we calculate the spatial relationship matrix by partitioning the spatial attention A_s into groups [45]. Specifically, the attention is divided into r groups, resulting in submatrices $A_{s1,g}$, $A_{s2,g}$ of size $r \times H \times (W/r)$. The grouping operation not only reduces computational complexity and makes each submatrix easier to process, but also enhances the model's ability to model local features and better capture the interaction relationships between bitemporal spatial features.

Subsequently, the spatial relationship affinity matrix $R_s \in \mathbb{R}^{B\times r\times r}$ is obtained by matrix multiplication between $A_{s1,g}$ and $A_{s2,g}$ as

$$R_s = A_{s1,g} \cdot (A_{s2,g})^T. \quad (2)$$

Next, define two learnable parameters α_1 and α_2, combine a fully connected (FC) layer and a sigmoid function (σ), and the weighted coefficients M_{s1} and M_{s2} are computed for the input features x_1 and x_2 as

$$M_{si} = \sigma(As + \alpha_i \cdot FC(\hat{R}_s)), i \in \{1, 2\} \quad (3)$$

where $\hat{R}_s$ denotes the flatten R_s.

Finally, the input features x_1 and x_2 are fused with the weights to produce the spatial relationship features $S_{\text{out}} \in \mathbb{R}^{B\times C\times H\times W}$, represented as

$$S_{\text{out}} = M_{s1} \cdot x_1 + M_{s2} \cdot x_2. \quad (4)$$

The grouped relationship matrix and weighted coefficients enable the model to strengthen the relationship modeling of spatial information while reducing computational complexity, making the output features more accurate in reflecting the change information in the spatial dimension.

2) Channel Relationship Modeling: For channel relationship modeling, the channel attention $A_c \in \mathbb{R}^{B\times C\times 1\times 1}$ is calculated as follows:

$$A_c = \sigma(\text{FC}(\text{GAP}(x))) \quad (5)$$

where GAP(·) represents global average pooling, and FC(·) represents a FC layer.

Similarly, the channel attention is divided into r groups, resulting in submatrices of size $B \times r \times (C/r)$, and the channel relationship is computed to obtain the resulting features C_{out}. By grouping the channel attention, the model can focus more on the feature dependencies within each group of channels, effectively capturing association information between different channels.

Finally, the spatial relationship features and channel relationship features are fused additively to obtain the final output features

$$F_{\text{out}} = S_{\text{out}} + C_{\text{out}}. \quad (6)$$

To fully leverage the advantages of both spatial and channel relationship modeling, we integrate these two types of features in a complementary manner. Spatial relationship modeling allows the model to focus on important spatial features, while channel relationship modeling better captures the interdependence between different channels. Through the mutual fusion of spatial and channel relationships, the TRAM effectively captures the spatial and channel relationships between bitemporal input features, achieving a more full feature relation interaction.

C. Higher Order Spatiotemporal Interaction Module

Remote sensing image change detection requires effectively perceiving semantic difference information between image pairs, which is high-order and complex, while preserving rich contextual features. Existing methods usually lack the ability to effectively model higher order feature interactions. To this end, we design a HOSTIM. This module introduces a higher order feature interaction mechanism [46], which recursively enhances the fine-grained feature relationship by feature division and effectively captures spatiotemporal dependencies of complex changes.

The HOSTIM consists of four consecutively stacked HOSTI blocks [47], denoted as HOSTIi, where $i \in \{1, 2, 3, 4\}$. Each HOSTI block contains two convolutional layers and two higher order interaction units (HOIUs). The structure of HOIU is shown in Fig. 3. HOIU is composed of the following key parts.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:07:29 UTC from IEEE Xplore. Restrictions apply.

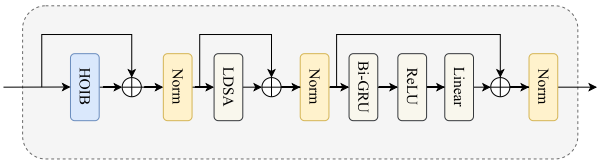


Fig. 3. Structure of HOIU. The HOIU consists of a HOIB, an LDSA, and a Bi-GRU, along with residual connections and layer normalization.

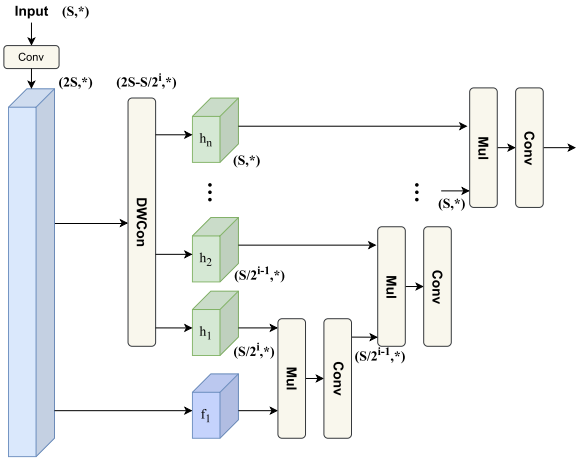


Fig. 4. Structure of HOIB. Composed of multiple stacked submodules, HOIB recursively refines spatial features while capturing complex high-order spatiotemporal dependencies.

1) HOIB is responsible for modeling complex higher order interaction relationships. The structure of HOIB is shown in Fig. 4.
2) Local dense synthesizer attention (LDSA) [48] enhances the expression capability of local features.

In addition, each HOIU includes residual connections and normalization operations, followed by a bidirectional GRU (Bi-GRU) to enhance sequence modeling capabilities after the HOIB and LDSA.

The main goal of the HOIB is to achieve the higher order spatial (S) interaction along temporal (T) and channel (C) dimensions through recursive operations. Each HOIB can be viewed as a recursive process that incrementally increases the interaction order of spatial features. Specifically, the input feature $X \in \mathbb{R}^{S\times T\times C}$ first undergoes 2-D convolution to expand the spatial dimension, generating the expanded feature $X' \in \mathbb{R}^{2S\times T\times C}$. Next, X' is divided into two spatial parts $f_1 \in \mathbb{R}^{S/2^i\times T\times C}$ and $h \in \mathbb{R}^{(2S-S/2^i)\times T\times C}$, where i represents the ith HOSTI. The f_1 retains part of the original feature information, while the h' is used to capture rich contextual information. Then, deep convolution is applied to the feature h' to obtain the contextual feature h

$$h = \text{DWConv}(h'). \tag{7}$$

Next, the contextual feature h is further divided into $n = i+1$ subfeatures along the spatial dimension as

$$h_1, h_2, \ldots, h_n = \text{Split}(h \cdot \alpha) \tag{8}$$

where α is a scaling coefficient used to stabilize training. As the number of recursive layers increases, the spatial dimension of the main feature f_1 decreases exponentially, and the spatial dimension of each subfeature h_k also decreases accordingly, while the spatial dimension of the contextual feature h increases to facilitate wider contextual interaction modeling.

Each layer of the recursion updates the features, which is implemented by the following recursive formula:

$$f_{k+1} = B_k(f_k) \odot h_k \tag{9}$$

where B is the kth$(1 \le k \le n)$ linear transformation used to match the spatial dimensions, and $\odot$ denotes elementwise multiplication. With each recursive iteration, the interaction order is incremented by 1. Such recursive updating allows the module to model arbitrary order spatiotemporal interactions.

In this way, the HOIB can recursively model the complex neighborhood relationships along the T and C dimensions while accumulating the interaction information between higher order features. Finally, the recursively generated feature f_{n+1} is output through 2-D convolution, completing the modeling process of the HOIB.

D. Loss Function

To enhance the model's optimization ability across different feature levels, we introduce a multistage supervision mechanism [49]. Different from the usual methods, in our model, deep supervision is applied to the decoder outputs, the encoder outputs of the CNN backbone, and HOSTIM. This ensures that feature learning at different stages is effectively and fully guided.

Specifically, we optimize the outputs of the convolutional encoder and HOSTIM by weighted binary cross-entropy (BCE) loss L_{wBCE} and weighted intersection over union (IoU) loss L_{wIoU} [50]. The two losses both assign higher weights to hard pixels, forcing the model to focus more on regions that are difficult to distinguish during training, so as that improve the model's detection accuracy in these areas. The combination of these two losses (namely, structure loss) ensures that the proposed model is fully optimized for local (pixel-level) restriction and high-order (global structure) restriction.

For the decoder output, the standard cross-entropy loss L_{CE} is applied to measure the difference between predicted results of RHighNet and true labels. The final total loss function is defined as

$$L_{\text{total}} = \alpha \cdot L_{\text{CE}} + \beta_1 \cdot L_{\text{wBCE}} + \beta_2 \cdot L_{\text{wIoU}} \tag{10}$$

where α is the weight for cross-entropy loss of decoder output, β_1 is the weight for local loss of convolutional encoder, and β_2 is the weight for global loss of HOSTIM. With this weighted loss function design, the model can effectively balance pixel-level accuracy and global feature interactions, ultimately achieving optimal change detection performance.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:07:29 UTC from IEEE Xplore. Restrictions apply.

IV. Experiments

A. Datasets

To validate the effectiveness of our proposed model, we conduct extensive experiments on four widely adopted benchmark datasets in the field of remote sensing change detection.

The LEVIR-CD dataset [51], a foundational resource for building change detection, features 637 high-resolution image pairs (10 241 024 pixels, 0.5 m/pixel) with 31 333 manually labeled building alteration instances. Chen et al. designed this dataset to facilitate algorithm development through a hierarchical partitioning scheme: 445 training pairs, 64 validation pairs, and 128 test pairs, each split into 256 256 nonoverlapping patches, resulting in 7120, 1024, and 2048 patches for respective subsets. The LEVIR-CD dataset can be used to train and evaluate building change detection algorithms, helping researchers to develop more effective models for identifying changes in building additions and deletions.

The WHU-CD dataset [52] captures post-earthquake urban reconstruction dynamics with two large-scale aerial images (3 250 715 354 pixels, 0.3 m/pixel) acquired in 2012 and 2016. After removing static patches, it generates 7120, 1024, and 2048 patch pairs for training, validation, and testing. This dataset is focused on large-scale building changes and can be used to assess the performance of the model in detecting large-scale building changes, which is particularly important in post-disaster reconstruction monitoring.

The SYSU-CD dataset [53] presents a diverse urban change scenario with 20 000 image pairs (256256 pixels) spanning 2007C2014 in Hong Kong. Covering construction, suburban expansion, vegetation shifts, and offshore activities, it partitions data into 12 000 training, 4000 validation, and 4000 test pairs. With its broad range of urban and environmental changes, this dataset is instrumental in developing models that generalize well across different types of landscape transformations, making it an essential resource for change detection applications.

The synthetic images with real season-varying RSCD dataset (CDD) [54] introduces 16 000 synthetic image pairs (256 256 pixels) to simulate seasonal and anthropogenic changes. Comprising 10 000 training, 3000 evaluation, and 3000 test pairs, CDD enables controlled experimentation on mixed-change scenarios (e.g., vegetation growth and urban development). This dataset can be used to evaluate the performance of models in dealing with seasonal and anthropogenic changes, and is particularly suitable for investigating the impact of seasonal changes on change detection algorithms.

B. Implementation Detail

The framework is implemented by the PyTorch programming environment, and the experiments are conducted on an Ubuntu system, equipped with an NVIDIA GeForce RTX 4090 GPU. RHighNet and other comparison methods are trained on a single GPU with a batch size of 16. The AdamW optimizer is employed, with a learning rate of 0.00009567 and a maximum of 200 epochs for training. In terms of data augmentation, flipping, cropping, blurring, and color transformations are applied.

TABLE I
Quantitative Evaluations on the LEVIR-CD Dataset

Method	Precision(%)	Recall(%)	F1(%)	IoU(%)	OA(%)
FC-EF [10]	86.91	80.17	83.42	72.01	97.29
IFNet [55]	89.74	85.26	87.34	78.23	98.70
STANet [51]	90.53	84.68	87.51	77.79	98.22
CDNet [56]	89.80	87.06	88.41	79.23	98.84
BIT [57]	92.67	87.61	89.32	80.72	99.00
ChangeFormer [58]	90.63	87.92	89.26	80.60	98.92
SNUNet [59]	90.89	88.31	89.28	80.55	98.93
P2V [60]	89.67	**91.78**	90.71	83.00	99.04
SEIFNet [61]	92.49	89.46	90.95	83.40	99.09
MFINet [62]	92.09	89.02	90.12	82.33	99.02
SAF [63]	91.60	88.70	89.43	81.66	98.95
RHighNet (ours)	**92.72** (+0.05)	89.94 (-1.84)	**91.31** (+0.36)	**84.01** (+0.61)	**99.13** (+0.04)

TABLE II
Quantitative Evaluations on the WHU-CD Dataset

Method	Precision(%)	Recall(%)	F1(%)	IoU(%)	OA(%)
FC-EF [10]	67.09	68.15	67.61	51.07	97.35
IFNet [55]	**95.72**	75.97	84.71	73.47	98.88
STANet [51]	83.81	**91.00**	87.26	77.40	98.66
SNUNet [59]	84.62	82.46	83.52	71.71	98.68
CDNet [56]	92.47	85.65	88.93	80.07	99.15
ChangeFormer [58]	91.71	90.29	90.99	83.48	99.28
P2V [60]	89.15	88.66	88.90	80.02	99.12
SEIFNet [61]	87.01	85.77	86.39	76.04	98.90
ICIF-Net [39]	92.98	85.56	88.32	79.24	98.96
RHighNet (ours)	93.97 (-1.75)	88.55 (-2.45)	**91.18** (+0.19)	**83.79** (+0.31)	**99.32** (+0.04)

C. Evaluation Metrics

In the experimental evaluation, we utilize several common metrics to measure the model's performance: precision, recall, overall accuracy, $F1$-score, and intersection over union (IoU). These metrics are widely used in classification and segmentation tasks to provide a comprehensive assessment of the model's effectiveness. Their calculation formulas are as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{11}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{12}$$

$$\text{Overall Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{13}$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{14}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{15}$$

where TP, TN, FP, and FN represent the numbers of true positives, true negatives, false positives, and false negatives, respectively. In all tables, the best results are in bold.

D. Quantify Analysis and Visualize Results

We conduct a comprehensive experimental evaluation on the aforementioned four datasets. The comparison algorithms include FC-EF [10], IFNet [55], STANet [51], CDNet [56], BiT [57], ChangeFormer [58], SNUNet [59], P2V [60], SEIFNet [61], ICIF-Net [39], MFINet [62], SAF [63], ISNet [64], DMINet [28], and MSCANet [65]. The quantitative

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:07:29 UTC from IEEE Xplore. Restrictions apply.

TABLE III
QUANTITATIVE EVALUATIONS ON THE SYSU-CD DATASET

Method	Precision(%)	Recall(%)	F1(%)	IoU(%)	OA(%)
FC-EF [10]	**91.30**	55.60	69.10	52.80	89.20
SNUNet [59]	84.57	74.66	79.31	65.71	90.81
CDNet [56]	76.54	**82.68**	78.49	65.96	89.94
BIT [57]	81.80	71.60	76.40	61.76	90.40
ChangeFormer [58]	86.75	72.92	79.24	65.62	90.99
ICIF-Net [39]	76.90	80.20	78.50	64.63	91.10
ISNet [64]	79.20	73.50	76.20	61.61	90.40
P2V [60]	82.17	74.36	78.07	64.03	90.15
SEIFNet [61]	84.90	74.10	79.10	65.47	91.30
RHighNet (ours)	85.24 (-6.06)	76.48 (-6.20)	**80.62** (+1.31)	**67.53** (+1.57)	**91.33** (+0.03)

TABLE IV
QUANTITATIVE EVALUATIONS ON THE CDD DATASET

Method	Precision(%)	Recall(%)	F1(%)	IoU(%)	OA(%)
FC-EF [10]	81.73	53.36	64.57	47.67	92.90
IFNet [55]	92.84	83.41	87.87	78.37	97.90
STANet [51]	92.26	84.49	88.20	78.90	97.84
SNUNet [59]	**97.68**	96.40	97.04	94.24	99.24
CDNet [56]	92.85	85.32	88.93	80.06	97.25
BIT [57]	92.63	83.91	88.06	78.66	97.84
ChangeFormer [58]	95.75	92.05	93.87	88.44	98.51
SEIFNet [61]	95.58	92.63	94.08	88.82	98.63
DMINet [28]	91.66	85.53	88.49	79.36	96.92
MSCANet [65]	92.11	84.43	88.10	78.74	97.85
SAF [63]	97.67	85.32	94.06	89.48	73.36
RHighNet (ours)	97.01 (-0.67)	**97.49** (+1.09)	**97.25** (+0.21)	**94.65** (+0.41)	**99.35** (+0.11)

TABLE V
EFFICIENCY COMPARISON OF DIFFERENT MODEL

Method	Flops (G)	Param (M)	F1(%)	IoU(%)
FC-EF [10]	**1.19**	**1.35**	83.42	72.01
IFNet [55]	50.71	82.26	87.34	78.23
STANet [51]	18.03	16.94	87.51	77.79
SNUNet [59]	43.94	12.03	89.28	80.55
BIT [57]	25.92	11.99	89.32	80.72
SAF [63]	14.47	40.22	89.43	81.66
MFINet [62]	6.89	4.95	90.12	82.33
Ours	66.51	96.81	**91.31**	**84.01**

evaluation results on the four datasets are presented in Tables I–IV, where the best results are bold, the second-best ones are underlined, and (.) denotes the performance improvement compared with the second-best results.

In the experiments on the four datasets, our method achieves superior performance. On the LEVIR-CD dataset, our method produces an $F1$-score of 91.31 and an IoU of 84.01, significantly outperforming other comparison algorithms. Compared with BIT that achieves an $F1$-score of 89.32, our method improves the $F1$-score by 1.99%. Existing methods, such as FC-EF and IFNet, exhibit certain limitations in feature processing and fusion, which hinder their ability to fully capture spatiotemporal change information from bitemporal remote sensing images. STANet and SNUNet methods improve feature representation to some extent. However, they still struggle to capture fine-grained details of change regions, leading to missed or false detections in complex scenarios. In contrast, our method excels in spatiotemporal feature modeling and relationship interaction, enabling it to effectively capture the spatiotemporal change information from bitemporal remote sensing images. On the WHU-CD dataset, our method achieves an $F1$-score of 91.18 and an IoU of 83.79, outperforming existing methods. For example, ChangeFormer achieves an $F1$-score of 90.99 and an IoU of 83.48, while CDNet reaches 88.93 and 80.07, respectively. ICIF-Net only yields an $F1$-score of 88.32 and an IoU of 79.24. In contrast, our method demonstrates superior performance in suppressing pseudo-changes. By incorporating the TRAM and HOSTIM, our model can filter out irrelevant background changes, effectively reducing the occurrence of pseudo-changes. For the SYSU-CD dataset, our method achieves an $F1$-score of 79.93 and an IoU of 66.57, which outperform the best-second results by 1.31% and 1.57%, respectively. This dataset features a variety of change types, including urban construction and vegetation changes. Previous methods often fail to capture fine-grained changes, especially in complex urban expansion or road construction scenarios. BIT improves feature representation by a transformer, only achieving an $F1$-score of 76.40 and an IoU of 61.76. The advanced ISNet method reaches an $F1$-score of 76.20 and an IoU of 61.61, which fails to capture more changes. Our model maintains high robustness across multiple change types through rich feature interactions. On the CDD dataset, our method achieves the best $F1$-score of 96.89 and IoU of 93.96, delivering excellent performance. Although various existing methods have made progress, there are still limitations in capturing fine-grained changes. SEIFNet enhances spatiotemporal features, but it lacks effective relationship interaction, achieving an IoU and $F1$-score of 88.82 and 94.08, respectively. By integrating both spatial and channel relationships, the proposed RHighNet can detect more subtle changes. In addition, the HOSTIM further refines the complex relationships between spatiotemporal features through a recursive high-order interaction mechanism, ensuring the model captures complex change information while retaining rich contextual details. It can also be seen from Table V that our model achieves the best performance with a limited increase in computational complexity.

The visualization results are shown in Figs. 5–8, where TP are represented by white pixels, TN by black pixels, FP by green pixels, and FN by red pixels. The comparison of detection results on different images shows the superiority of our method. In the LEVIR-CD dataset, although the change regions are relatively small, our method accurately captures these subtle changes, demonstrating its ability to detect fine spatiotemporal changes. However, other methods struggle to detect small change regions accurately, often missing subtle changes. This results in an increased number of false positives and false negatives, which lowers overall detection accuracy. The improvement of our RHighNet can be attributed to the TRAM and HOSTIM, which enhance the model's sensitivity to minor changes and suppress pseudo-changes, especially in complex background scenarios. In the WHU-CD dataset, existing methods tend to misidentify background as a change region and often fail to distinguish real changes from complex

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:07:29 UTC from IEEE Xplore. Restrictions apply.

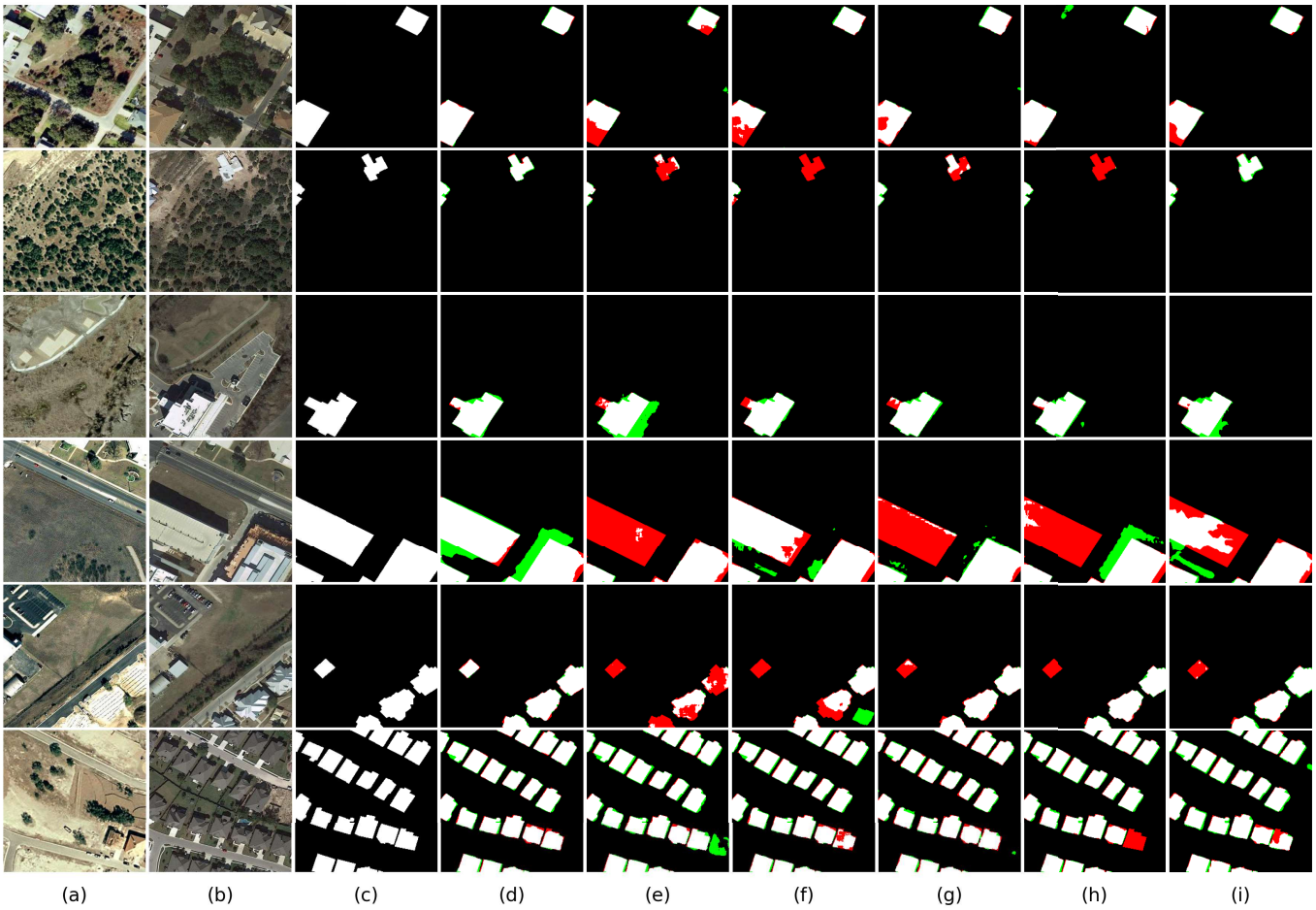


Fig. 5. Visualization results on the LEVIR-CD dataset. (a) T_1 images. (b) T_2 images. (c) Ground truth. (d) Proposed RHighNet. (e) CDNet. (f) SNUNet. (g) ChangeFormer. (h) P2V. (i) SEIFNet.

TABLE VI
ABLUTION STUDY IN IOU INDEX

Methods				Datasets			
	HOSTIM	TRAM	Structure Loss	LEVIR-CD	WHU-CD	SYSU-CD	CDD
Baseline				81.29	81.22	64.19	92.33
Baseline	✓			82.01	82.67	65.18	93.12
Baseline	✓	✓		83.32	82.81	67.18	94.35
Baseline	✓	✓	✓	**84.01**	**83.79**	**67.53**	**94.65**

background, leading to an increased number of false positives. In contrast, our method filters out background interference and focuses solely on the true change regions to a certain extent, thus improving detection accuracy. In the SYSU-CD dataset, there are many large, smooth areas of change, which easily cause missed detection for difficult areas. It can be seen that most comparison methods tend to overlook some change regions, leading to a higher false-negative rate. Comparatively, our method consistently identifies the change areas and reduces missed detection. This improvement is largely due to exploring multiple relationships between spatiotemporal features. In the CDD dataset, other methods often produce blurred edges. Our method ensures that the edges of change regions are clearly delineated, and the integrity of change areas is better preserved. Therefore, our method enables more accurate identification of change regions.

E. Ablation Study

To validate the effectiveness of different modules in our proposed framework, we conduct ablation experiments on four datasets. Table VI presents the quantitative results of the ablation experiments.

1) Influence of the HOSTIM: In change detection tasks, traditional convolutions often fail to effectively model complex contextual interactions, leading to insufficient detection of change regions, while the multihead self-attention (MHSA) mechanism in Transformer can model global dependencies

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:07:29 UTC from IEEE Xplore. Restrictions apply.

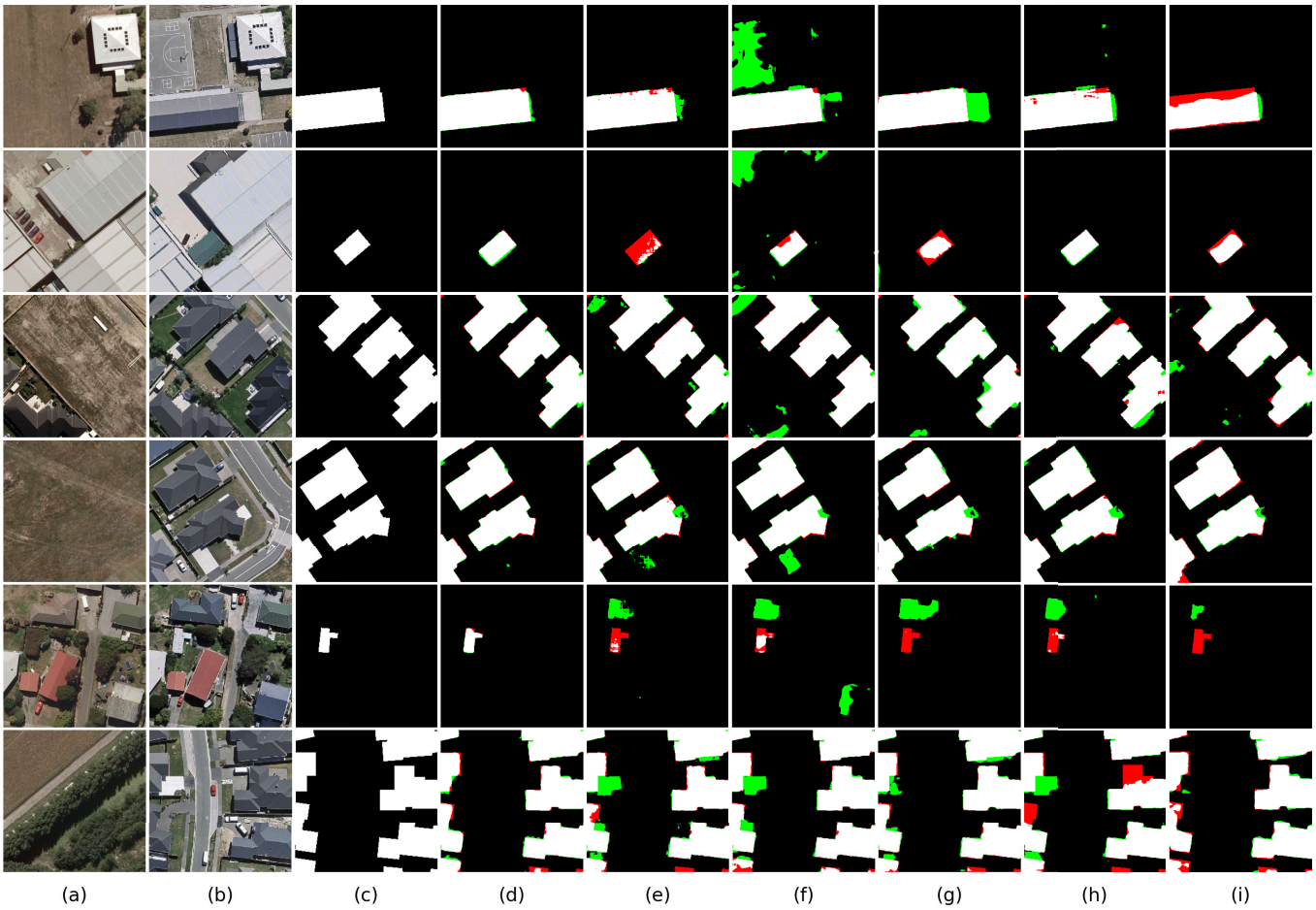


Fig. 6. Visualization results on the WHU-CD dataset. (a) T_1 images. (b) T_2 images. (c) Ground truth. (d) Proposed RHighNet. (e) CDNet. (f) SNUNet. (g) ChangeFormer. (h) P2V. (i) SEIFNet.

but suffers from quadratic complexity $O(N^2)$, resulting in high computational costs and limiting its application in large-scale remote sensing images. The proposed HOSTIM can model higher order information interaction with lower computational complexity through recursive feature modeling, which enhances the ability to capture complex change patterns. It can be seen, the IoU scores obtained by adding this module are 82.01% (an increase of 0.72%), 82.67% (an increase of 0.83%), 65.18% (an increase of 0.99%), and 93.12% (an increase of 0.79%) on the LEVIR-CD, the WHU-CD, the SYSU, and the CDD, respectively. The improvements across all datasets demonstrate the effectiveness of the HOSTIM in capturing complex changed patterns by higher order interactions in bitemporal features.

2) Influence of the TRAM: Existing methods often rely on simple pixel difference or channel concatenation operations to fuse bitemporal features, which makes it difficult to better model the spatial and channel relationships between bitemporal features, leading to feature confusion and accuracy degradation. The proposed TRAM improves the relational modeling ability of bitemporal features by calculating the spatial and channel attention interaction of bitemporal features, which can collaboratively focus on important changed relationships. The quantitative results with this module show the IoU scores on the four datasets further improve to 83.32% (an increase of 1.31%), 82.81% (an increase of 0.14%), 67.18% (an increase of 2.00%), and 94.35% (an increase of 1.02%) on the LEVIR-CD, the WHU-CD, the SYSU-CD, and the CDD, respectively. This indicates the TRAM significantly improves the model's ability to better reveal the changing relationship between bitemporal features, and thus gains the detection accuracy.

To further improve interpretability, we also visualize the attention distributions of the TRAM and HOSTIM in Fig. 9. It can be observed that, as the depth of the network increases, the TRAM gradually focuses from the detailed information at the lower level to the semantic change-related area at the higher level between bitemporal images. By incorporating spatial and channelwise relationship modeling, the module enhances localized interaction awareness through a grouped attention mechanism. In contrast, the HOSTIM exhibits a more concentrated response in change regions. Its core higher order interaction modeling mechanism, through spatial recursive partitioning and contextual enhancement operations, gradually strengthens the modeling of global spatiotemporal dependencies among features, leading to more precise attention distribution and clearer background regions.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:07:29 UTC from IEEE Xplore. Restrictions apply.

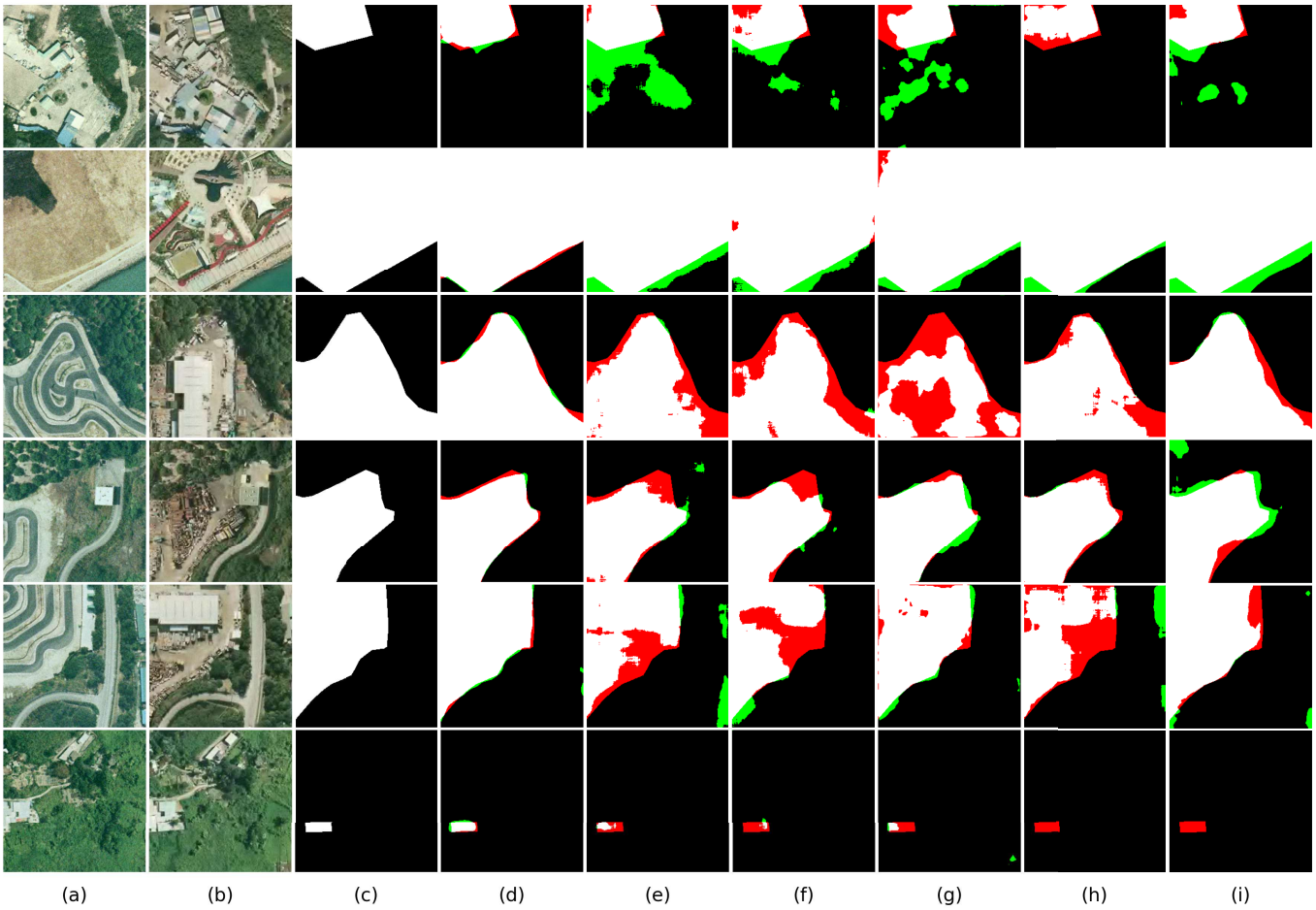


Fig. 7. Visualization results on the SYSU-CD dataset. (a) T_1 images. (b) T_2 images. (c) Ground truth. (d) Proposed RHighNet. (e) CDNet. (f) SNUNet. (g) ChangeFormer. (h) P2V. (i) SEIFNet.

3) Influence of Loss Function: Most existing methods rely exclusively on cross-entropy loss or BCE loss to optimize the model. This often leads to suboptimal accuracy in the change detection task. It can be seen from Table VI, the multistage loss with structure loss witnesses a boost. And, its accuracy curve shown in Fig. 10 is overall more stable, and the accuracy is significantly higher than the model without structure loss in most of the training epochs. Specifically, the IoU values on the four datasets all increase, which are 0.69%, 0.98%, 0.35%, and 0.3%, respectively, indicating the effectiveness of this approach. This also demonstrates that the introduction of deep supervision at different feature levels empowers the model to learn more robust and context-aware representations. These representations allow the model to gain a more comprehensive understanding of the changes occurring in remotely sensed image pairs. In essence, structure loss acts as a catalyst, driving the model to extract richer and more discriminative features, thereby significantly improving overall performance in the change detection task.

F. Parameter Analysis

1) Group Number in TRAM: In the change detection task, it is essential to improve detection performance for accurately modeling the spatiotemporal relationships between bitemporal input features. The TRAM enables better fusion of bitemporal image features by modeling the spatial and channel relationships between them, thereby enhancing the model's ability to capture change regions. A critical hyperparameter in TRAM is the number of groups, which determines how spatial attention is partitioned and directly influences the performance of feature fusion. To investigate the effect of this parameter, we conduct a parameter analysis study on the LEVIR-CD dataset.

TABLE VII
IOU AND $F1$-SCORES FOR DIFFERENT NUMBERS OF GROUPS

Number of Groups	IoU (%)	F1 (%)
8	83.67	91.11
16	83.73	91.14
32	**84.01**	**91.31**
64	83.29	90.88

The results presented in Table VII show significant variation in IoU and $F1$ performance as the number of groups increases. With lower group values (8 and 16), the model achieves IoU scores of 83.67% and 83.73%, respectively. At these values,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:07:29 UTC from IEEE Xplore. Restrictions apply.

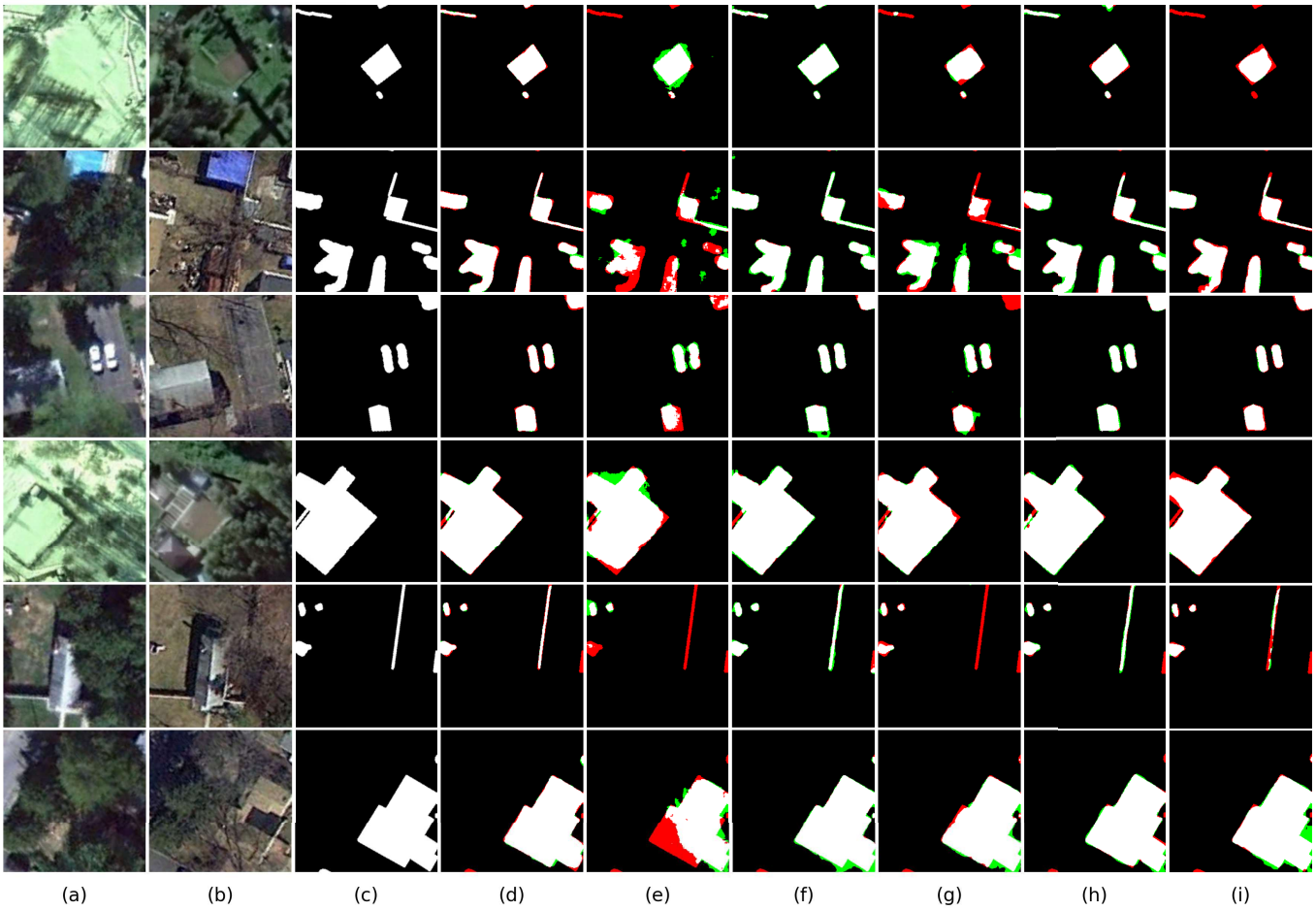


Fig. 8. Visualization results on the CDD dataset. (a) T_1 images. (b) T_2 images. (c) Ground truth. (d) Proposed RHighNet. (e) CDNet. (f) SNUNet. (g) ChangeFormer. (h) P2V. (i) SEIFNet.

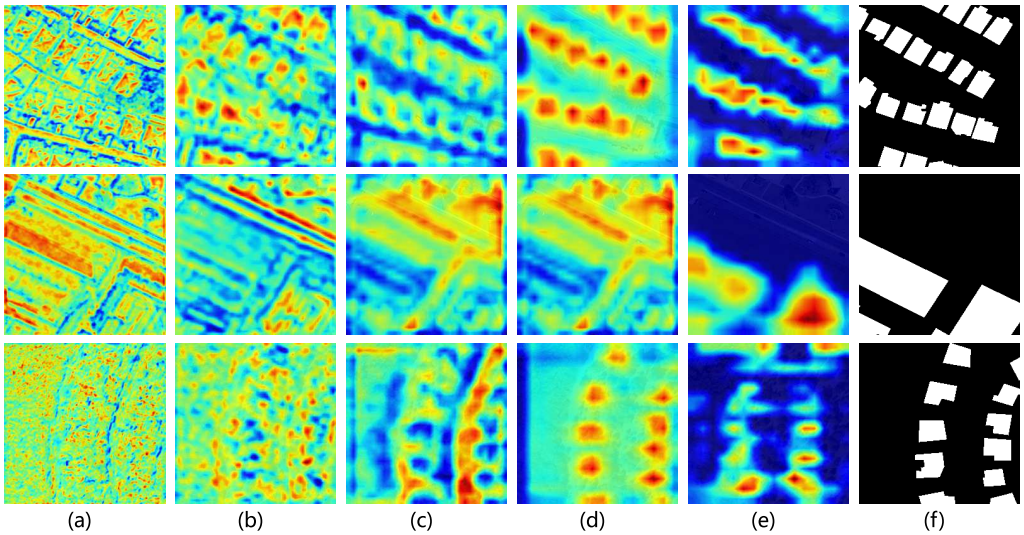


Fig. 9. Visualization results of the attention weights of HOSTIM and TRAM. (a)–(d) Attention heatmaps of TRAM at different scales. (e) Attention heatmap of HOSTIM. (f) Ground truth.

the insufficient partitioning of features hinders the model from effectively modeling the complex relationships between features, resulting in suboptimal feature fusion performance. However, when the number of groups is increased to 32, the IoU improves substantially to 84.01%, reaching the best performance. This improvement suggests that a greater number

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:07:29 UTC from IEEE Xplore. Restrictions apply.

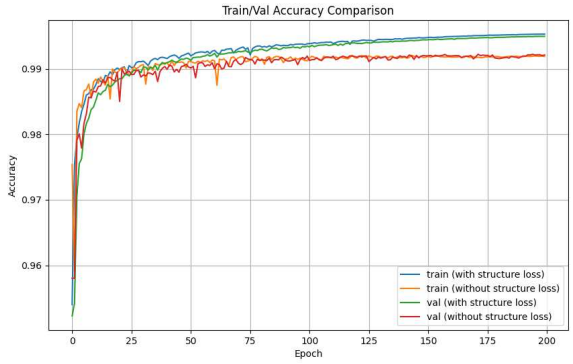


Fig. 10. Training curve of multistage loss on training and validation accuracy.

TABLE VIII
IoU Results With Different Loss Weight Settings on Four Datasets

α, β_1, β_2	LEVIR-CD	WHU-CD	SYSU-CD	CDD
0.5, 0.3, 0.2	82.28	82.27	65.20	93.12
0.6, 0.2, 0.2	**84.01**	**83.79**	66.57	93.96
0.7, 0.15, 0.15	83.65	82.83	67.03	94.15
0.7, 0.2, 0.1	83.72	82.72	**67.53**	**94.65**
0.7, 0.1, 0.2	83.59	82.56	66.90	94.07

of groups allows the model to refine feature representations more effectively, enhancing the modeling of relationships across spatial and channel dimensions, thus improving the detection accuracy.

Conversely, when the number of groups is further increased to 64, the IoU slightly decreases to 83.29%. The higher group number results in an excessive partitioning of feature information, making the model confused to effectively focus on key information during the fusion process, which negatively impacts its performance. This highlights the importance of carefully selecting the number of groups in TRAM. An optimal group number can significantly enhance model performance; too many groups may lead to information fragmentation and confusion. In our experiments, 32 groups in TRAM are set to achieve the optimal performance.

2) Loss Weight: To study the effect of different weights for loss, we conduct the analysis experiments with varied weight settings of the loss function across the four datasets. As shown in Table VIII, the setting of (0.6, 0.2, 0.2) performs best on the LEVIR-CD and the WHU-CD, where foreground–background distribution is relatively balanced. The low-level structure's detailed features may help with precise recognition. In contrast, the SYSU-CD and CDD datasets have more imbalanced changes with frequent occlusions and significant illumination variations, where high-level semantic features with distinctiveness are more important. The setting of (0.7, 0.2, 0.1) achieves the best performance on the two datasets. These results empirically validate that appropriate weight between decoder loss and structure supervision helps the model to adapt to specific dataset characteristics such as object scale, occlusion, and illumination changes, so that it achieves better detection performance.

G. Network Structure Analysis

1) Structure of the Backbone: In order to explore feature fusion structures that are more suitable for change detection, we conduct comparative experiments with different backbone structures. Three different structures are designed.

1) Without TRAM, where the two input features (x and y) of main branch are not processed by TRAM.
2) Both main branches are processed through TRAM, where the fusion results of TRAM are respectively fused with the original features and propagated to the next layer.
3) A side structure (as shown in Fig. 1), where TRAM fusion results are input only to the x branch, while the y side continues to use the original input features for forward propagation. The comparison results are shown in the left part of Fig. 11.

For the first structure, although the decoder receives the fused features, the absence of relationship modeling between bitemporal input features limits the model's ability to recognize changes accurately, resulting in the IoU of only 83.12%. In the second structure, it effectively captures the relationships between input features, improving the model's ability to identify change regions, resulting in an IoU of 83.71%. By combining feature fusion with the original features, the model can more comprehensively understand the change relationships between spatiotemporal features, which enhances its sensitivity to change regions. However, the addition of features may introduce redundancy, potentially affecting the fusion effectiveness. In the third structure, it performs the best. The x side retains the original input features without undergoing feature fusion. This differentiation allows the y side to fully benefit from TRAM, enhancing its ability to capture semantic change features in image pairs, while the x side maintains the original features, avoiding potential information loss or increased complexity from excessive fusion. By adopting this differentiated treatment, the third structure optimizes computational efficiency while maximizing the model's ability to identify change regions.

2) Structure of the TRAM: Given the functional distinctions between channel and spatial attention, the order in which they are applied may influence the overall performance of the model. Channel attention typically captures dependencies between channels at a global level, while spatial attention focuses on the interrelationships of features within local regions. Consequently, the sequence in which these attentions are applied can determine how information from each is fused, thus affecting the final performance. Therefore, we compare three different configurations for combining channel attention and spatial attention: 1) channel first, followed by spatial (CS); 2) spatial first, followed by channel (SC); and 3) parallel addition fusion (PAF).

The experimental results in the right part of Fig. 11 demonstrate that PAF significantly enhances the model performance. Compared with serial processing, the PAF simultaneously models both spatial and channel relationships, which enables more effective integration of multidimensional features, yielding superior performance in complex changing scenes. In

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:07:29 UTC from IEEE Xplore. Restrictions apply.

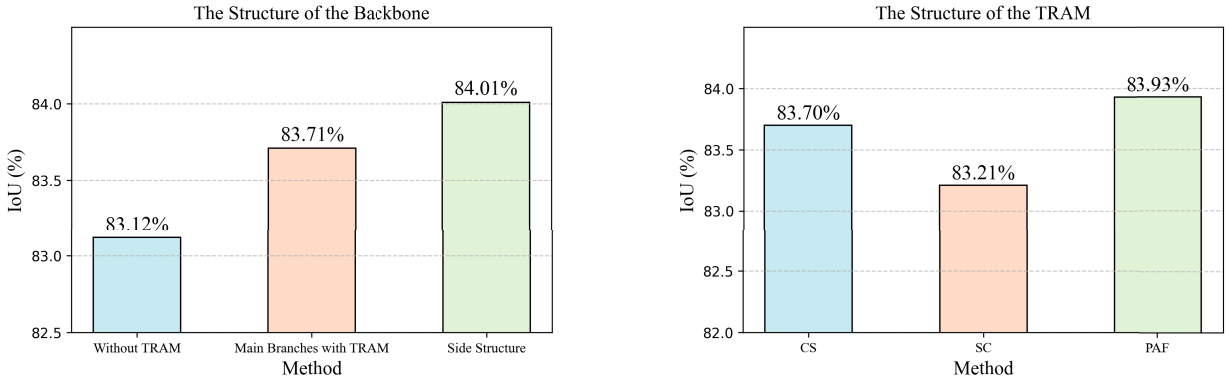


Fig. 11. Network structure analysis.

serial processing, spatial and channel features are handled separately, optimizing each individually and failing to fully capture their interdependencies, which results in slightly lower performance. In contrast, parallel fusion not only preserves the independence of channel and spatial features but also captures their interdependencies, allowing the model to gain a more comprehensive understanding of spatiotemporal features. As a result, the PAF performs better and can improve the accuracy and robustness of the model in detecting changes.

V. Conclusion

In this article, we propose the RHighNet, which features two key modules, TRAM and HOSTIM. The TRAM effectively models the spatial and channel relationships between bitemporal images, enabling the fusion of temporal features across various scales and capturing the interdependencies between these features. The HOSTIM adopts a recursive higher order feature interaction mechanism, which enables features to gradually accumulate more complex contextual dependencies in spatiotemporal dimensions through high-order feature transformations. Thus, complex neighborhood relationships and spatiotemporal changes can be modeled more effectively. In addition, a multistage loss is introduced for local pixel-level restriction and global structure restriction, which helps to smooth training and improve prediction accuracy. Experimental results show that the proposed RHighNet outperforms existing state-of-the-art methods in the change detection task, providing more accurate and reliable detection performance. Future research will focus on exploring more advanced fusion and interaction strategies to further enhance the detection accuracy and generalization capabilities of the model. In addition, since the computational cost of RHighNet remains relatively high, we also consider developing lightweight design strategies with effectiveness.

References

[1] S. Kumar, S. Arya, and K. Jain, "A SWIR-based vegetation index for change detection in land cover using multi-temporal Landsat satellite dataset," *Int. J. Inf. Technol.*, vol. 14, no. 4, pp. 2035–2048, Jun. 2022.

[2] J. Li, Y. Pei, S. Zhao, R. Xiao, X. Sang, and C. Zhang, "A review of remote sensing for environmental monitoring in China," *Remote Sens.*, vol. 12, no. 7, p. 1130, Apr. 2020.

[3] P. de Bem, O. de Carvalho Junior, R. F. Guimarães, and R. T. Gomes, "Change detection of deforestation in the Brazilian Amazon using Landsat data and convolutional neural networks," *Remote Sens.*, vol. 12, no. 6, p. 901, Mar. 2020.

[4] K. Li, X. Cao, and D. Meng, "A new learning paradigm for foundation model-based remote-sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610112.

[5] W. A. Malila, "Change vector analysis: An approach for detecting forest changes with Landsat," in *Proc. LARS Symposia*, 1980, p. 385.

[6] J. Chen, P. Gong, C. He, R. Pu, and P. Shi, "Land-use/land-cover change detection using improved change-vector analysis," *Photogramm. Eng. Remote Sens.*, vol. 69, no. 4, pp. 369–379, Apr. 2003.

[7] P. Du, X. Wang, D. Chen, S. Liu, C. Lin, and Y. Meng, "An improved change detection approach using tri-temporal logic-verified change vector analysis," *ISPRS J. Photogramm. Remote Sens.*, vol. 161, pp. 278–293, Mar. 2020.

[8] J. S. Deng, K. Wang, Y. H. Deng, and G. J. Qi, "PCA-based land-use change detection and analysis using multitemporal and multisensor satellite data," *Int. J. Remote Sens.*, vol. 29, no. 16, pp. 4823–4838, Aug. 2008.

[9] A. A. Nielsen, "The regularized iteratively reweighted MAD method for change detection in multi- and hyperspectral data," *IEEE Trans. Image Process.*, vol. 16, no. 2, pp. 463–478, Feb. 2007.

[10] C. Wu, B. Du, and L. Zhang, "Slow feature analysis for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 5, pp. 2858–2874, May 2014.

[11] M. Gong, L. Su, M. Jia, and W. Chen, "Fuzzy clustering with a modified MRF energy function for change detection in synthetic aperture radar images," *IEEE Trans. Fuzzy Syst.*, vol. 22, no. 1, pp. 98–109, Feb. 2014.

[12] J. Zhou, Y. Liu, B. Peng, L. Liu, and X. Li, "MaDiNet: Mamba diffusion network for SAR target detection," *IEEE Trans. Circuits Syst. Video Technol.*, early access, May 28, 2025, doi: 10.1109/TCSVT.2025.3574657.

[13] Z. Lv, H. Huang, L. Gao, J. A. Benediktsson, M. Zhao, and C. Shi, "Simple multiscale UNet for change detection with heterogeneous remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[14] M. K. Panda, A. Sharma, V. Bajpai, B. N. Subudhi, V. Thangaraj, and V. Jakhetiya, "Encoder and decoder network with ResNet-50 and global average feature pooling for local change detection," *Comput. Vis. Image Understand.*, vol. 222, Sep. 2022, Art. no. 103501.

[15] S. Saha, L. Mou, X. X. Zhu, F. Bovolo, and L. Bruzzone, "Semisupervised change detection using graph convolutional network," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 4, pp. 607–611, Apr. 2021.

[16] H. Lyu, H. Lu, and L. Mou, "Learning a transferable change rule from a recurrent neural network for land cover change detection," *Remote Sens.*, vol. 8, no. 6, p. 506, Jun. 2016.

[17] W. Liu, Z. Kang, J. Liu, Y. Lin, Y. Yu, and J. Li, "A multitask CNN-transformer network for semantic change detection from bitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5647215.

[18] F. Yang et al., "ConvFormer-CD: Hybrid CNN-transformer with temporal attention for detecting changes in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5614815.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:07:29 UTC from IEEE Xplore. Restrictions apply.

[19] C. Zhang, L. Wang, and S. Cheng, "HCGNet: A hybrid change detection network based on CNN and GNN," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4401412.

[20] Z. Li, C. Yan, Y. Sun, and Q. Xin, "A densely attentive refinement network for change detection based on very-high-resolution bitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4409818.

[21] K. Li, Z. Li, and S. Fang, "Siamese NestedUNet networks for change detection of high resolution satellite image," in *Proc. 1st Int. Conf. Control, Robot. Intell. Syst.*, Oct. 2020, pp. 42–48.

[22] T. Xie, L. Fu, D. Zhao, Y. Hu, H. Xing, and S. Wang, "Object-level change detection via Siamese detection network," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2024, pp. 10372–10375.

[23] L. He et al., "Change-guided similarity pyramid network for semantic change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5637917.

[24] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4409720.

[25] H. Yu, H. Yang, L. Gao, J. Hu, A. Plaza, and B. Zhang, "Hyperspectral image change detection based on gated spectral–spatial–temporal attention network with spectral similarity filtering," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511313.

[26] X. Zhang, L. Wang, and S. Cheng, "A multiscale cascaded cross-attention hierarchical network for change detection on bitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 2002216.

[27] J. Chen et al., "DASNet: Dual attentive fully convolutional Siamese networks for change detection in high-resolution satellite images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 1194–1206, 2021.

[28] Y. Feng, J. Jiang, H. Xu, and J. Zheng, "Change detection on remote sensing images using dual-branch multilevel intertemporal network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401015.

[29] Y. Liu et al., "Building change detection in earthquake: A multiscale interaction network with offset calibration and a dataset," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5635217.

[30] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.

[31] C.-F. R. Chen, Q. Fan, and R. Panda, "CrossViT: Cross-attention multi-scale vision transformer for image classification," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 357–366.

[32] Y. Jiang et al., "PolarFormer: Multi-camera 3D object detection with polar transformer," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 1042–1050.

[33] X. Ma, X. Zhang, M.-O. Pun, and M. Liu, "A multilevel multimodal fusion transformer for remote sensing semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5403215.

[34] Z. Yang et al., "CogVideoX: Text-to-video diffusion models with an expert transformer," in *Proc. 13th Int. Conf. Learn. Represent.*, 2024, pp. 83048–83077.

[35] X. Song, Z. Hua, and J. Li, "Remote sensing image change detection transformer network based on dual-feature mixed attention," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5920416.

[36] X. Zhang, S. Cheng, L. Wang, and H. Li, "Asymmetric cross-attention hierarchical network based on CNN and transformer for bitemporal remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 2000415.

[37] W. Zhang, Y. Zhang, S. Gao, X. Lu, Y. Tang, and S. Liu, "Spectrum-induced transformer-based feature learning for multiple change detection in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5500212.

[38] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[39] Y. Feng, H. Xu, J. Jiang, H. Liu, and J. Zheng, "ICIF-Net: Intra-scale cross-interaction and inter-scale feature fusion network for bitemporal remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410213.

[40] Y. Wu et al., "CSTSUNet: A cross Swin transformer-based Siamese U-shape network for change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5623715.

[41] J. Zhao et al., "GeoFormer: A geometric representation transformer for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4410617.

[42] Z. Li et al., "STADE-CDNet: Spatial–temporal attention with difference enhancement-based network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611617.

[43] X. Zhao, S. Li, T. Geng, and X. Wang, "GTransCD: Graph transformer-guided multitemporal information united framework for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5500313.

[44] C. Shi, Z. Zhang, W. Zhang, C. Zhang, and Q. Xu, "Learning multiscale temporal–spatial–spectral features via a multipath convolutional LSTM neural network for change detection with hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5529816.

[45] S. Guo et al., "ISDNet: Integrating shallow and deep networks for efficient ultra-high resolution segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4351–4360.

[46] Y. Rao, W. Zhao, Y. Tang, J. Zhou, S.-N. Lim, and J. Lu, "HorNet: Efficient high-order spatial interactions with recursive gated convolutions," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 10353–10366.

[47] W. Wei, Y. Hu, H. Huang, and L. He, "IIFC-Net: A monaural speech enhancement network with high-order information interaction and feature calibration," *IEEE Signal Process. Lett.*, vol. 31, pp. 196–200, 2024.

[48] M. Xu, S. Li, and X.-L. Zhang, "Transformer-based end-to-end speech recognition with local dense synthesizer attention," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2021, pp. 5899–5903.

[49] D.-P. Fan et al., "PraNet: Parallel reverse attention network for polyp segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2020, pp. 263–273.

[50] A. Lin, B. Chen, J. Xu, Z. Zhang, G. Lu, and D. Zhang, "DS-TransUNet: Dual Swin transformer U-Net for medical image segmentation," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–15, 2022.

[51] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[52] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[53] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604816.

[54] M. A. Lebedev, Y. V. Vizilter, O. V. Vygolov, V. A. Knyaz, and A. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *Int. Arch. Photogramm.*, vol. 42, pp. 565–571, Jan. 2018.

[55] C. Zhang et al., "A deeply supervised image fusion network for change detection in high resolution bi-temporal remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 166, pp. 183–200, Aug. 2020.

[56] H. Chen, W. Li, and Z. Shi, "Adversarial instance augmentation for building change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5603216.

[57] H. Chen, Z. Qi, and Z. Shi, "Efficient transformer based method for remote sensing image change detection," 2021, *arXiv:2103.00208v1*.

[58] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[59] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 8007805.

[60] M. Lin, G. Yang, and H. Zhang, "Transition is a process: Pair-to-video change detection networks for very high resolution remote sensing images," *IEEE Trans. Image Process.*, vol. 32, pp. 57–71, 2023.

[61] Y. Huang, X. Li, Z. Du, and H. Shen, "Spatiotemporal enhancement and interlevel fusion network for remote sensing images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5609414.

[62] W. Ren, Z. Wang, M. Xia, and H. Lin, "MFINet: Multi-scale feature interaction network for change detection of high-resolution remote sensing images," *Remote Sens.*, vol. 16, no. 7, p. 1269, Apr. 2024.

[63] H. Yin, C. Ma, L. Weng, M. Xia, and H. Lin, "Bitemporal remote sensing image change detection network based on Siamese-attention feedback architecture," *Remote Sens.*, vol. 15, no. 17, p. 4186, Aug. 2023.

[64] G. Cheng, G. Wang, and J. Han, "ISNet: Towards improving separability for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5623811.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:07:29 UTC from IEEE Xplore. Restrictions apply.

[65] M. Liu, Z. Chai, H. Deng, and R. Liu, "A CNN-transformer network with multiscale context aggregation for fine-grained cropland change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4297–4306, 2022.



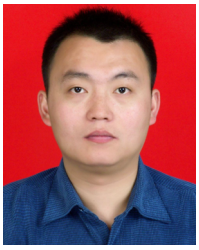
Huihui Dong received the B.E. degree in computer science and technology from Henan University, Kaifeng, China, in 2015, and the Ph.D. degree in electronic science and technology from Xidian University, Xi'an, China, in 2022.

She is currently a Tenured Associate Professor with the School of Information and Control Engineering, Xi'an University of Architecture and Technology, Xi'an. Her research interests include computer vision and remote sensing image understanding.



Xinyu Du received the B.E. degree in computer science and technology from Xi'an University of Architecture and Technology, Xi'an, China, where she is currently pursuing the M.E. degree in computer technology with the School of Information and Control Engineering.

Her research interests include remote sensing image change detection and deep learning.



Zhijie Li received the B.E. degree in computer science and technology, the M.E. degree in computer application technology, and the Ph.D. degree in digital architecture from Xi'an University of Architecture and Technology, Xi'an, China, in 2003, 2006, and 2015, respectively.

His research interests include digital architecture, pattern recognition, and image and graphics processing.

Dr. Li is currently a member of the Special Committee on Image and Graphics Engineering, Shaanxi Society of Electronics. He is also a member of China Society of Image and Graphics (CSIG), Shaanxi Society of Image and Graphics, and Shaanxi Society of Computer Education.



Zongfang Ma received the B.S. degree in communication engineering and the M.S. degree in technology for computer applications from Xi'an University of Architecture and Technology, Xi'an, China, in 2002 and 2006, respectively, and the Ph.D. degree in control science and engineering from Northwestern Polytechnical University, Xi'an, in 2011.

He is a Professor with the School of Information and Control Engineering, Xi'an University of Architecture and Technology. His research interests include intelligent information processing and industrial applications of machine vision.



Yumeng Wang received the Ph.D. degree in pattern recognition and intelligent systems from the Institute of Automation, Chinese Academy of Sciences, Beijing, China, in 2017.

She previously served as a Computer Vision Algorithm Engineer at the Intel China Research Center, Beijing. She is currently an Assistant Professor with the School of Information and Control Engineering, Xi'an University of Architecture and Technology, Xi'an, China. Her research focuses on computer vision, with particular emphasis on object detection and segmentation, video understanding, abnormal behavior detection, and prediction.



Hao Zhu (Member, IEEE) received the B.S. degree in physics and photoelectricity engineering and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, China, in 2013 and 2019, respectively.

He is currently an Associate Professor with the School of Artificial Intelligence and the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, Xidian University. His research interests include deep learning, multimodal remote sensing image fusion and classification, and remote sensing object detection and tracking.



Wenping Ma (Senior Member, IEEE) received the B.S. degree in computer science and technology and the Ph.D. degree in pattern recognition and intelligent systems from Xidian University, Xi'an, China, in 2003 and 2008, respectively.

She is currently a Professor with the School of Artificial Intelligence, Xidian University. Her research interests include natural computing and intelligent image processing.

Dr. Ma is a member of Chinese Institute of Electronics (CIE).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:07:29 UTC from IEEE Xplore. Restrictions apply.